

Kyle Main

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Data Engineer — Large-Scale Pipelines, Distributed Processing & Experimentation Analytics

Data engineer with 12 years of experience building the ingestion pipelines, canonical data models, and distributed processing systems that large organizations rely on for trustworthy analytics at scale. Built and ran ingestion pipelines for 220+ unique log sources, designed a Common Information Model standardizing field names/types across all of it, and used Apache Beam/GCP Dataflow, PySpark, and GCP Dataproc for large-scale data processing and exploratory analysis. Deep, hands-on statistical and data-science background — time-series analysis, clustering, and EDA at scale — directly relevant to building data infrastructure that has to hold up to statistical scrutiny.

CORE SKILLS

Data Engineering & Pipelines: Data pipeline design and ingestion for 220+ unique log sources; ETL/data normalization (50+ Logstash filters); Common Information Model / data dictionary design and standardization; Apache Beam / GCP Dataflow for large-scale historical data processing; connector/collector health monitoring and troubleshooting; Python, SQL

Distributed & Big Data Processing: PySpark, SparkSQL (writing, debugging, and optimizing Spark jobs); GCP Dataproc compute clusters; BigQuery; GCP serverless/event-driven data enrichment; working exposure to Kafka and Flink streaming jobs

Data Science / Statistics: Exploratory data analysis at scale (Zeppelin notebooks, home-grown analysis toolkit); time-series anomaly detection; clustering/unsupervised ML; Pandas, scikit-learn, NumPy, SciPy, PyTorch, R; quantitative/numerical modeling background (M.S. Physics)

Cloud & Engineering Tools: GCP, AWS, Azure; Git; GitLab CI/CD; Docker

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Built an entirely new data ingestion platform for DOE/NNSA from the ground up — CrowdStrike, Suricata, and Zeek telemetry into a central Elasticsearch environment — including custom data transforms, a behavioral-analytics layer built on top of those transforms, and data-quality monitoring/alerting content (DOE/NNSA Security Data Integration, completed).
- Supported data ingestion and data-quality engineering within an Elasticsearch/Splunk environment for DOE's Continuous Diagnostics and Mitigation (CDM) program, a large-scale federal continuous-monitoring data platform (completed).
- Currently builds and maintains analytics content (Splunk saved searches) supporting Treasury's Security Operations Center (Treasury SOC / TSSOC, current project).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and engineering team building signature, statistical, time-series, and ML-based analytics content against massive-scale customer data on cloud-based big-data platforms.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Owned data engineering for a next-gen cloud-based SIEM at scale: built ingestion pipelines for 220+ unique log sources, wrote 50+ Logstash parsing/normalization filters, and designed a Common Information Model — a data dictionary standardizing field names and types across all parsed data.
- Built a homegrown Apache Beam program run via GCP Dataflow to fetch large volumes of historical cold-storage data for customers; assisted building "Loggify," a homegrown log parsing/filtering tool that replaced Logstash; owned connector/collector health monitoring and troubleshooting for the ingestion pipeline.
- Ran exploratory data analysis at scale on GCP Dataproc compute clusters using Zeppelin notebooks and a home-grown reusable analysis toolkit, with PySpark/SparkSQL jobs to load and analyze data from cloud storage buckets.
- Applied statistical and ML techniques to large-scale telemetry — time-series anomaly detection (authentication behavior, process chains) and unsupervised clustering (device behavior) — and built/managed 2,300+ detection rules covering most of the MITRE ATT&CK matrix as a very early hire.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Analyzed large-scale security log datasets to build custom detection models — DNS-based malware detection/mitigation and anomalous-behavior discovery across network data.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox